

Chapter 7

Using Famous Convolutional Neural Networks (CNNs)

Objective

To understand the architecture of famous CNNs such as GoogLeNet and similar convolutional neural networks and implement them from scratch for image classification tasks on CIFAR-10. The focus will be on building these architectures, training them, and evaluating their performance.

1. Famous CNN Architectures Overview

Modern CNN architectures have been designed for improved efficiency and accuracy. Below are two notable ones:

1. GoogLeNet (Inception): Known for its inception modules, which use multiple filter sizes to capture features at various scales.
2. VGGNet: A simpler architecture using stacked convolutional layers of the same filter size to extract hierarchical features.

Tasks:

1. Research the architectural components of GoogLeNet and VGGNet.
2. Discuss how these architectures improve upon a basic CNN.
3. Explore the evolution of CNN architectures over time and how they influenced modern designs.

2. Designing Architectures from Scratch

The goal is to implement the architectures of GoogLeNet and VGGNet from scratch using TensorFlow/Keras Sequential API. You will train these architectures on CIFAR-10.

Tasks:

1. Research and define the architecture of GoogLeNet and VGGNet.
2. Implement the architectures from scratch based on your research.
3. Train the models on CIFAR-10 and record their performance metrics (e.g., training time, accuracy, and loss).
4. Document and visualize the architectural differences between GoogLeNet and VGGNet using diagrams.

3. Training and Evaluation

Train each architecture from scratch on CIFAR-10 and compare their performance.

Tasks:

1. Train GoogLeNet and VGGNet for 20 epochs with a batch size of 64.
2. Evaluate their accuracy and loss on the test set.
3. Compare the results and training times of the two architectures.
4. Experiment with different learning rates and optimizers. Report how these changes affect performance.

4. Data Augmentation

Data augmentation can improve model generalization by increasing the diversity of the training data.

Tasks:

1. Apply common data augmentation techniques (e.g., flipping, rotation, brightness adjustments) to the CIFAR-10 dataset.
2. Train the models with augmented data and compare the results to the original models.
3. Discuss the impact of data augmentation on training time and accuracy.

5. Visualizing Feature Maps

Understanding how CNNs learn can be enhanced by visualizing the outputs of intermediate layers (feature maps).

Tasks:

1. Select a few test images and pass them through your trained models.
2. Visualize the feature maps at different layers of GoogLeNet and VGGNet.
3. Compare the patterns and features captured by earlier layers versus deeper layers.
4. Discuss how the visualized features contribute to classification.

6. Reflection

Questions to Reflect On:

1. How do the performance metrics (accuracy, loss) of GoogLeNet and VGGNet compare?
2. What challenges did you face while implementing these architectures from scratch?
3. How does training from scratch differ from using pretrained models?
4. How did data augmentation influence the training and results?
5. What insights did you gain from visualizing feature maps?

Additional Resources

1. Papers:

- GoogLeNet: "Going Deeper with Convolutions"
- VGGNet: "Very Deep Convolutional Networks for Large-Scale Image Recognition"